MAMI Human Milk Microbiome — Compositional Analysis Phase

CLR/Aitchison analysis, zero-replacement sensitivity and study-associated feature screening

Data scope. Analysed 1,128 MAMI breast-milk samples and 344 operational feature labels after excluding only the explicit Mammalia, Chloroplast and Mitochondria columns. The uploaded table contains relative-abundance-like values, not raw sequencing counts. No clinical covariates or negative-control profiles were available in the uploaded files.

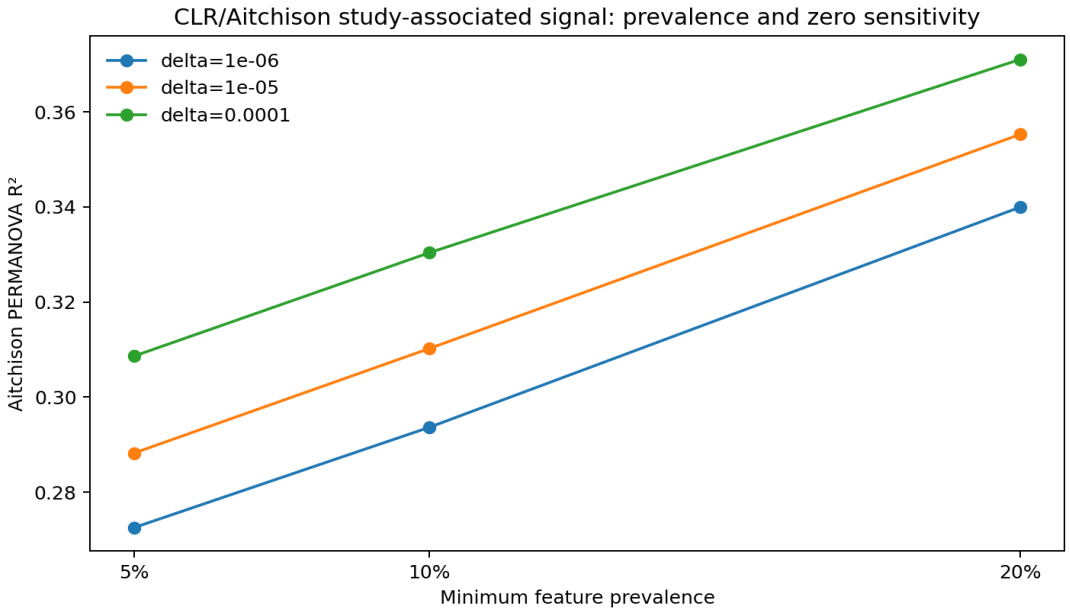
1. Why a compositional approach?

Relative abundances are constrained to a total and therefore cannot be interpreted as independent absolute quantities. This phase uses centred log-ratio (CLR) transformations and Aitchison (Euclidean CLR) distances. Because many values are zero, each analysis replaces zeros with one of three small values (1e-6, 1e-5 or 1e-4 in the closed relative-abundance scale), re-closes each sample, and then applies CLR. This is a sensitivity analysis, not a claim that any chosen pseudocount is biologically correct.

2. Community-level study association

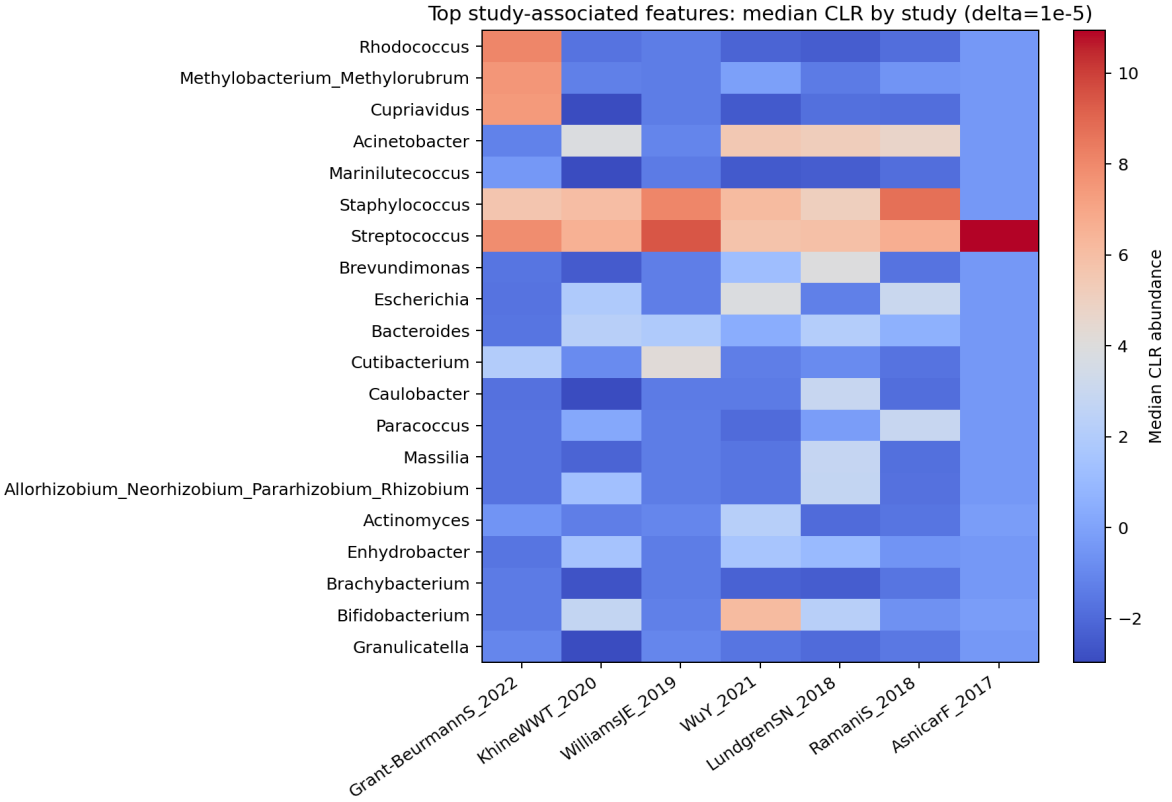
Aitchison PERMANOVA and a within-group dispersion diagnostic were repeated for 5%, 10% and 20% feature prevalence cutoffs, and all three zero-replacement values. Results are tabulated below; the plot summarizes PERMANOVA R^2.

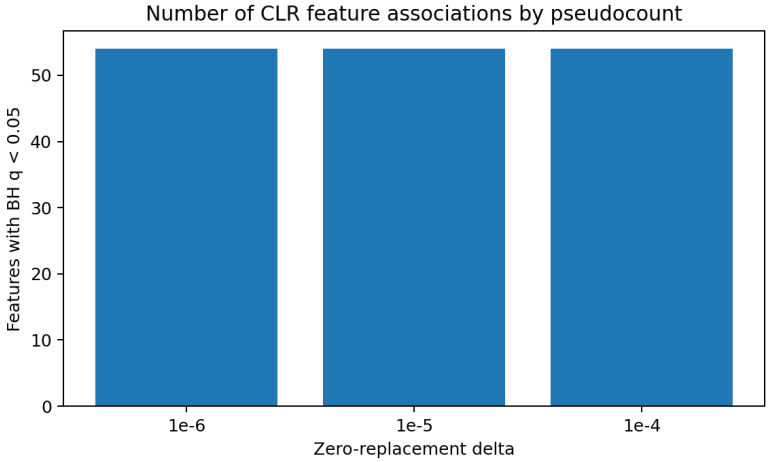
Min prev.	Features	Delta	Aitchison R^2	PERMANOVA p	PERMDISP p
5%	100	1e-06	0.273	0.001	0.001
5%	100	1e-05	0.288	0.001	0.001
5%	100	0.0001	0.309	0.001	0.001
10%	54	1e-06	0.294	0.001	0.001
10%	54	1e-05	0.310	0.001	0.001
10%	54	0.0001	0.330	0.001	0.001
20%	27	1e-06	0.340	0.001	0.001
20%	27	1e-05	0.355	0.001	0.001
20%	27	0.0001	0.371	0.001	0.001



3. Feature-level CLR screen

Among features present in at least 10% of samples (n=54), Kruskal–Wallis tests compared CLR values across the seven studies, with Benjamini–Hochberg false-discovery correction. With delta=1e-5, 54 features had q<0.05. Across all three zero-replacement settings, 54 features remained q<0.05 at every setting; 49 also had the same study with the highest median CLR at all three settings.





Feature	Prev.	H	Epsilon²	BH q	Highest median CLR
Rhodococcus	51%	862.0	0.764	3.4e-181	Grant-BeurmannS_2022
Methylobacterium_Methylorubrum	63%	775.1	0.686	1.0e-162	Grant-BeurmannS_2022
Cupriavidus	42%	566.5	0.500	6.9e-118	Grant-BeurmannS_2022
Acinetobacter	57%	465.7	0.410	2.7e-96	WuY_2021
Marinilutecoccus	24%	419.2	0.369	2.3e-86	AsnicarF_2017
Staphylococcus	91%	399.9	0.351	2.6e-82	RamaniS_2018
Streptococcus	95%	353.1	0.310	2.5e-72	AsnicarF_2017
Brevundimonas	22%	350.6	0.307	7.6e-72	LundgrenSN_2018
Escherichia	25%	349.1	0.306	1.4e-71	WuY_2021

Feature	Prev.	H	Epsilon²	BH q	Highest median CLR
Bacteroides	35%	323.6	0.283	3.8e-66	KhineWWT_2020
Cutibacterium	51%	314.9	0.276	2.6e-64	WilliamsJE_2019
Caulobacter	11%	306.2	0.268	1.7e-62	LundgrenSN_2018

4. Interpretation and limits

CLR results are relative to the geometric mean of the included features. A positive CLR does not mean a taxon has higher absolute biomass; it means it is high relative to the sample geometric mean. The Kruskal–Wallis tests identify differences among study distributions, not cause or country-specific effects. Country, study, sequencing region/protocol, laboratory workflow and sample processing remain confounded. Unequal group sizes and significant dispersion tests further limit simple interpretation.

The feature screen is exploratory. The uploaded feature table does not provide raw read counts, sequencing depths, raw reads, full taxonomy lineages, extraction blanks or negative controls. For these reasons, this analysis does not claim to perform a definitive ANCOM-BC2/DESeq2 analysis, does not identify contaminants, and does not establish biologically meaningful differential abundance. The zero-replacement sensitivity helps reveal dependence on a modelling choice, but does not remove these limitations.

5. What is required for manuscript-grade validation

Obtain raw sequence reads or the original unrarefied feature-count table; exact ASV/OTU sequence and full taxonomy assignment with classifier/database versions; extraction blanks and PCR/library negative controls; positive controls if available; study-specific primer/variable-region, extraction, amplification and processing protocols; and participant-level metadata with clear unique subject identifiers. Then repeat using a validated compositional differential-abundance method appropriate to the count input (for example, ANCOM-BC2 on suitable count data), with study-aware design and sensitivity analyses.

Exploratory statistical analysis only. No clinical or causal conclusions.